

Adham Ahmed

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Experience

halan (First private unicorn in Egypt) Jun. 2024 – Present

Software Engineer

Go, Scala, MongoDB, Kafka, Kubernetes, Redis, Elasticsearch

- Assumed technical ownership of the **Halan Investments** platform, managing backend systems that hold over **3 Trillion EGP** in equity funds while resolving critical production incidents.
- Dramatically optimized a highly concurrent service, **increasing throughput by 50x per pod**. This eliminated the need for expensive horizontal scaling by enabling the concurrent processing of 50 requests within a single pod.
- Architected a robust, cross-microservice flow for processing trading orders, ensuring fault-tolerant integration with third-party systems like the **Egyptian Bourse**.
- Pioneered a **Server-Driven UI (SDUI)** architecture, drastically reducing mobile release cycles by engineering the backend to dynamically render native iOS and Android components.
- Built the **Secured Loan Lock** mechanism, a complex feature safely locking a user's equity funds to leverage their investments as proof of income to secure credit balances.
- Managed the full technical lifecycle for new product initiatives, translating business requirements into system design, delegating tasks across engineering teams, and overseeing successful production rollouts.
- Modernized the investment cash-in flow by integrating it with Halan's **Unified Payment Gateway (UPG)**, driving the initiative through staging, rigorous regression testing, and deployment.
- Played an instrumental role in the successful production launch of the **Halan Wallet**, navigating and implementing stringent technical and regulatory requirements for **ABK Bank**.
- Engineered the **Microfinance Profile** module, leading to a **40% increase** in users viewing their installment plans and an **88% increase** in in-app installment payments.
- Contributed to a major migration of the loan and installment payment service from Scala to Go, transitioning to an event-driven model using **Kafka** to ensure reliable communication with the core banking system.

bld.ai Nov. 2022 - Nov. 2023

Software Engineer

Python, Django, Git, GitLab, Open-AI, React.js, Docker

- Worked on an internal landing page project at bld.ai, utilizing ChatGPT API to generate dynamic content from Google Sheets data.
- Collaborated across teams to align project objectives, enhancing user engagement.

bld.ai Jul. 2022 - Oct. 2022

Software Engineering Intern

Python, JavaScript, HTML, CSS, Bootstrap, Django, React.js

- Completed a full-stack internship at bld.ai, gaining proficiency in frontend development with React.js and backend development using Django and REST APIs.

Orange Innovation Nov. 2021 - May. 2022

Software Engineering Intern

Java, SQL, Spring, Spring Boot, Threads

- Contributed to the development of a production project involving a chatbot deployed on Smartly.ai, built using Spring Boot for the backend. The chatbot engaged users by suggesting healthy food recipes and provided YouTube videos for the chosen recipe YummyBot.
- Learned Java sockets, JDBC, and threads.

Education

October 6 University (O6U) Sep. 2019 - Jul. 2023

B.S. in Computer Science

Achieved an Excellent Degree for my graduation project.

Projects

E-learning Management System (ELMS) Feb. 2023 - Ongoing

E-learning Website Meeting, Assignments & Quizzes

Python, JavaScript, Django, DRF & channels, React.js

- Developed a student-to-student connectivity feature using Django Channels.
- Integrated Google Calendar API to automate event scheduling and notifications for platform users.
- Implemented features for assignment submission and quizzes within the platform.

Biometric Attendance	Oct. 2023 - Ongoing
<i>Desktop Application project to create reports and Calculate salaries.</i>	<i>Python, Tkinter, OpenPyXL</i>
• Desktop application with a user interface to collect employee data and attendance events from the ZK-14 Biometric Machine. • Generate attendance and payroll reports for employees in a company.	

Honors, Awards and Competitions

- Participated at Arab African Collegiate Programming Contest (2022)
- Participated at Egyptian Collegiate Programming Contest (2021, 2022, 2023)
- Solved +2000 problems on various online judges

Leadership and Volunteering

ICPC O6U Community	Sep. 2020 - present
<i>Co-founder & Mentor</i>	
• Guiding a community of CS students in their problem solving and Competitive Programming skills. • Orchestrating coding sessions, and coding competition events to enhance skills and encourage continuous learning.	

Skills

Languages:

Python, Java, C/C++, SQL, HTML, JavaScript

Technologies & Tools:

Django, React.js, Git, Linux, Docker, CI/CD (familiar with), Spring & Spring Boot

concepts:

Object Oriented Programming, Problem Solving, Algorithms, Data Structures, Testing.